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Weeding out or picking winners in open innovation? Factors driving multi-stage crowd selection on LEGO ideas

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ABSTRACT

Using unique data from the LEGO Ideas platform and a novel approach of algorithm-based abduction, we combine multiple methods to provide new insights into crowd selection. Through qualitative content-coding, interviews, and prior literature, we derive an initial set of variables. We then use machine learning for feature selection and to identify the most important factors for crowd selection. The findings are used to build theory on crowd selection, which we test on a hold-out sample. Our key finding is that ideator and idea characteristics suggested by prior research can predict crowd selection only at early stages. More specifically, crowds rely on ideator status, prior success, and a carefully crafted idea presentation with many images to weed out bad ideas in early stages. However, these characteristics have little bearing in predicting winners at later stages. We explain this with signaling theory, where these ideator and idea characteristics represent strong, costly signals for idea quality and value only in early stages but fade as new signals (such as social signals of popularity and gaining fact traction) emerge. Our study provides two main contributions to research on crowd selection and idea evaluation. First, our approach enables us to prune explanations for crowd selection and guide attention to the factors which matter most. Second, we extend prior work by considering multi-stage crowd selection and highlighting its dynamic nature.

1. Introduction

Only 15 hours after Xin Jun submitted her idea "Spongebob Squarepants – The Krusty Krab" to LEGO Ideas, it reached 100 votes from the crowd. While happy, she was not entirely surprised her idea passed the first stage of crowd selection. After all, she chose a popular franchise and put in a lot of care to develop the presentation, making it vibrant, colorful, and attention-grabbing. And the work paid off. 1000 votes after 4 days, 5000 after 23 days. After only 43 days, her idea reached the ultimate goal: 10,000 votes. And all that without any serious promotion efforts. Xin could not really explain her success. Sure, she had the presentation, the topic, and the timing. But other ideas had that as well and still got stuck.[2]

We use unique, in-depth data from LEGO Ideas to understand better how crowds select ideas. LEGO Ideas is an interactive platform where users like Xin can submit ideas for new LEGO sets and comment and vote on each other's ideas. Ideas must pass multiple stages of crowd selection to become an official set. LEGO Ideas is one of the premier examples of successful crowdsourcing. The platform has produced several commercial hits and is a central pillar in the company's efforts to engage with its customers and fans.[3]

In line with LEGO's experiences, open innovation research highlights

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[1] Independent researcher.
[2] This vignette is based on the interview we did with Xin on December 2, 2020. To ensure our account matched her perception, we asked Xin to review the vignette text and she approved it.
[3] We have no financial interest in LEGO. We signed an NDA to access the data, but we were allowed to disclose the company name for this publication. Data access took several years and had to comply with GDPR.

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the potential of crowds to generate ideas for new products and services (Dahlander et al., 2021; Faraj et al., 2016; Jeppesen and Frederiksen, 2006; Zaggl et al., 2023), which can be more novel than those generated by employees (Poetz and Schreier, 2012). At the same time, this research suggests that organizations find it challenging to process a large number of ideas (Blohm et al., 2018; Terwiesch and Xu, 2008) and identify the few worth implementing (Piezunka and Dahlander, 2015). Crowds' abilities to *generate ideas* scale quicker than organizations' abilities to *select*, which is problematic because organizations can miss out on novel ideas (Piezunka and Dahlander, 2015), and the many necessary rejections could damage their relationship with the crowd members (Piezunka and Dahlander, 2019). One possible solution is using crowds to generate *and* select ideas (e.g., Fuchs et al., 2010; Hoornaert et al., 2017), just like LEGO.

In crowd selection, organizations empower crowds to evaluate ideas through a voting system—so-called crowd voting (Toubia and Florès, 2007). Crowd voting helps the organization collect diverse viewpoints, which can collectively indicate ideas' market potential (Kornish and Ulrich, 2014; Magnusson et al., 2016) and reduce the cognitive burden of selecting ideas for internal experts by limiting the choice set (Blohm et al., 2016). Despite the growing reliance on crowds for idea selection, Csaszar (2018, p. 278) points out that "the field of organizations does not offer guidance on when and how to use crowds" for idea selection. It is still an open question *how* crowds select ideas and how this selection process changes throughout the idea contest (Csaszar, 2018; Mollick and Nanda, 2016). Understanding which factors are responsible for crowd selection helps scholars focus attention on the most important factors with the highest predictive power and organizations to find markers of what constitutes success when a new idea pops up.

In addition, many organizations design crowd selection as a sequential multi-stage process. For instance, like LEGO Ideas, the crowdsourcing platforms *99designs* and *IBM Innovation Jam* involve crowds in multiple rounds of voting to weed out bad ideas and identify the few final winners. Prior work so far has focused on single-stage settings, investigating how certain idea and ideator characteristics predict crowd selection (see e.g. Piezunka and Dahlander, 2015; Riedl and Seidel, 2018; Hoornaert et al., 2017). Nevertheless, multi-stage selection processes differ from single-stage settings in their design. They represent a dynamic environment where new signals and information become available to the crowd and the ideators as ideas progress through selection stages. Ideas also gain and lose momentum and as new ideas are continuously added, they compete with each other and within a certain timeline, like in the case of LEGO Ideas. Given its different design, it is unclear whether the insights generated from prior research extend to multi-stage selection. Factors that affect crowd selection at early stages may differ from those influential at later stages. It is thus important to develop theory to explain selection dynamics in multi-stage settings.

Therefore, we ask: how does crowd selection differ in early vs later stages of the selection process? Considering the limited knowledge of how crowds select ideas and the fragmented literature with competing explanations, we resort to an abductive approach. Following Shrestha et al. (2021) and He et al. (2020), we combine machine learning (ML) with qualitative insights to build theory on crowd selection. Our research approach consists of four steps. First, we had several conversations with experts from LEGO, content-coded a subset of ideas, and consulted the literature to specify an initial set of variables that explain crowd selection. Second, we used ML for feature selection to detect robust patterns in the data. Third, we built theory based on these patterns, existing literature, and our qualitative data. Fourth, we used regression models to test the theory on a hold-out sample.

We make two contributions to the literature on crowd selection and idea evaluation. First, our approach guides attention to the factors with the greatest explanatory power, which means we can prune explanations with little importance. Prior work focuses on single (or a few) idea and ideator characteristics that affect crowd selection (see, e.g., Piezunka and Dahlander, 2015; Riedl and Seidel, 2018; Hoornaert et al., 2017). While this is important and common in our field, it opens the possibility that we, as a community of scholars, add more and more explanations when a simpler story may suffice. Just like pruning allows for new growth of a tree by removing dead or dying branches, pruning explanations and building more parsimonious theory can spur new insights on crowd selection.

Second, we extend research on crowd selection by considering multi-stage processes, which is important because differences in platform design affect crowds' contributions and behaviors (Barbosu and Gans, 2022). Building on signaling theory (Spence, 1973), we shed light on the role of signals for crowd selection and their dynamic effects across selection stages, in line with recent calls (Moradi et al., 2023). Our main finding is that some of the factors proposed by prior work (such as ideator status, ideator experience, or idea presentation) only explain crowd selection at early stages, but conditional on passing that threshold, they cannot predict which idea ultimately wins. We theorize that these ideator- and idea-related factors constitute valuable, costly signals only in early stages of crowd selection. They help to weed out low-quality ideas. However, as ideas progress to later selection stages and new signals become available, these early signals fade and become less effective (Xia et al., 2016). Passing earlier stages of crowd selection is a strong and visible signal of popularity and community approval, which can trigger social influence processes and increase ideators' promotion efforts. Underlining the dynamic nature of multi-stage crowd selection, we also find that gaining fast traction is a key but often overlooked success factor, which can be a powerful signal of idea quality at later stages. As crowds pay more attention to these new signals, earlier signals become less useful for predicting winners in the final stage. Overall, by distinguishing between pattern detection and pattern explanation (He et al., 2020), our algorithm-based abductive approach enables us to prune explanations as well as generate novel insights on crowd selection.

2. Literature review

2.1. Drivers of crowd selection

A key aspect of understanding crowd selection is what underlies crowd support (Franke et al., 2014). In general, crowd support emerges because crowd members (a) express their perception of quality, (b) express their preferences, and (c) are subject to social influence. Creativity research highlights that evaluating creative ideas is inherently related to assessing their *quality*, typically evaluated in terms of idea novelty and usefulness (Zhou et al., 2019). Through voting, crowds express their beliefs about the best ideas. The votes are based on crowd members' subjective assessments (Cattani and Ferriani, 2008; Zhou et al., 2017) of how novel and useful the ideas are to them. Studies have found that crowds' evaluations tend to conform to those of experts when it comes to selecting high-quality ideas (Magnusson et al., 2016; Toubia and Florès, 2007; Mollick and Nanda, 2016), with crowds even being more open toward novel ideas than experts (Berg, 2016). Crowd selection is thus an attractive and useful complement to companies' selection, enabling organizations to identify promising ideas.

At the same time, crowd support indicates crowd members' *preferences* (Askin and Mauskapf, 2017). Through voting, crowds express their preferences concerning the ideas they want implemented (Dahlander et al., 2019). Crowdsourcing platforms often consist of users of the firm's products who know their needs and wants best (Magnusson et al., 2014). Crowd voting thus expresses collective preferences that indicate the market potential of ideas (Di Gangi and Wasko, 2009; Jensen et al., 2014). It is often challenging to disentangle whether quality perceptions or preferences underlie crowd support, as they typically exist in parallel.

Finally, *social influence* can drive crowd support (Frey and van de Rijt, 2020; Keuschnigg and Ganser, 2016). It can shape beliefs about idea quality and potentially even preferences. When faced with uncertainty about an idea's value, crowds follow their own views *and* the

choices and opinions of others to make selection decisions (Smirnova et al., 2022). This is the case especially when individuals can observe others' voting, which affects crowds' evaluations of ideas and shapes their quality perceptions (Aral and Walker, 2014). Crowd voting reveals the popularity of an idea, and crowd members may use this social information as a signal of idea quality, causing herding effects (Muchnik et al., 2013). Such herding can be driven by 'irrational' behaviors (Mollick and Nanda, 2016), where crowd members imitate others. As social influence increases, predicting outcomes becomes more complex, and various outcomes become possible above a certain quality threshold (Salganik et al., 2006).

2.2. Crowd selection in single-stage settings

While part of the literature discusses how crowd support emerges, some studies have investigated how specific factors related to (a) the ideator and (b) the idea affect crowd selection in single-stage settings. Certain ideator and idea characteristics increase the chances of crowd selection because they reflect the learning and effort typically needed for high-quality submissions. Moreover, crowds may use these characteristics as signals of idea quality and value—an aspect we will return to later in our theorizing.

First, when it comes to *ideator characteristics*, prior work has mainly focused on how ideator experience and status affect crowd selection. An ideator's experience and expertise reflect competence and knowledge and can translate into future support (Bayus, 2013; Deichmann and van den Ende, 2014; Li et al., 2016). Ideators with greater platform tenure or who have proposed multiple ideas learn how to improve subsequent ideas from first-hand experience; they can also learn indirectly by supporting and discussing others' ideas (Riedl and Seidel, 2018). Individuals with a record of past success are more likely to have their ideas selected by the crowd in the future (Deichmann and van den Ende, 2014). Prior experiences of success can trigger learning, as ideators learn which ideas and idea presentations the crowd prefers (Deichmann and van den Ende, 2014). Moreover, some studies show that an ideator's status in the community is important for attracting crowd support (Szatmari et al., 2021). Status reflects the community's recognition and is accrued through high-quality contributions and commitment (Smirnova et al., 2022).

Second, the *idea characteristics* themselves affect crowd selection (Zhou et al., 2019). Idea characteristics are usually expressed in unstructured, written text (Hoornaert et al., 2017). Fueled by new computational methods, researchers have developed measures to assess how specific characteristics of the idea presentation can predict an idea's potential. For instance, some studies focus on the idea description, showing that ideas with more elaborated and understandable descriptions are perceived to be of higher quality and are thus more likely to be selected (Jensen et al., 2014; Hoornaert et al., 2017). Other studies show that ideas with more positive descriptions, more images, and more tags are more likely to receive crowd support (Mollick, 2014; Courtney et al., 2017).

Prior work often focuses on single (or a few) ideator and idea characteristics to explain crowd selection. While such in-depth investigations are important, the downside is that the literature becomes fragmented: we end up with various potential explanations for why crowds select ideas without knowing which ones matter the most. Moreover, our understanding of crowd selection is limited because evidence thus far comes exclusively from single-stage contests, and it is unclear whether the findings extend to multi-stage processes.

2.3. Crowd selection in multi-stage selection

While prior work focuses on single-stage settings, some organizations design multiple-stage crowd selection. For example, some firms use several voting rounds, with only the highest-ranked ideas advancing to the next stage. Others—like LEGO—require ideas to achieve a certain level of crowd support within a particular timeframe to qualify for the next stage of the competition. The advantages of such multi-stage processes are that they filter out bad ideas at early stages, where ideas are plentiful (Klein and Garcia, 2015), and that the convergence toward a narrower set of ideas indicates their market and creative potential (Huang et al., 2014).

Multi-stage selection processes differ from single-stage processes on several important dimensions. First, multi-stage selection implies multiple rounds of voting, and selection into subsequent stages is contingent upon the outcome of prior stages. This means new signals emerge as the ideas progress through selection stages, reaching more and new crowd members. Passing earlier stages of crowd selection are strong signals. Crowd members can observe how ideas pass the different selection stages, giving them strong social signals of popularity and community approval. Second, multi-stage processes are typically longer, and ideas gain and lose momentum. The selection deadlines and thresholds can introduce particular dynamics that do not exist in single-stage contests. On LEGO Ideas, for instance, ideators often increase their promotion efforts when time is running out to reach the next selection stage. Finally, as ideas progress through the selection stages, they could be perceived as more valuable by the crowd *and* the ideators. Like job candidates in advanced job talks or researchers in later rounds of the review process, ideators feel closer to their goals. As a result, they become more motivated to invest time and effort to attract further crowd support and see their ideas win. For all these reasons, factors that drive crowd selection in multi-stage contests may differ from those in single-stage settings. Crowds may also place weight on different factors across selection stages so that influential factors at early stages potentially differ from those at later stages. Distinguishing single- and multi-stage crowd selection is also important because recent work has highlighted that differences in platform design affect crowds' contributions and behaviors (Barbosu and Gans, 2022).

3. Analytical strategy

Given the fragmented literature on crowd selection and the differences between single- and multi-stage processes, simply dropping prior theoretical ideas into a hypothesis framework is not warranted. Instead, we rely on an algorithm-based abductive approach (Bamberger, 2018; Shrestha et al., 2021 for an empirical example) to generate new insights into how crowds select ideas at different selection stages. This also allows us to identify additional factors that affect crowd selection but that prior work has not considered. We build theory using algorithm-supported abduction (Shrestha et al., 2021; He et al., 2020) and combine qualitative methods with machine learning and regression-based hypotheses testing.

We follow a four-step research design that we illustrate in Fig. 1. In Step 1, we conducted interviews, visits, and observations to understand the research phenomenon and context. We also derived qualitative insights from content-coding of 160 ideas at different stages of selection as well as consulted relevant literature to identify an initial set of variables that could explain crowd selection. In Step 2, we used different ML algorithms to establish robust associations between this initial set of variables and crowd selection at different stages. In Step 3, we used abductive reasoning to explain the ML and qualitative findings and develop a novel set of hypotheses on (multi-stage) crowd selection. Finally, in Step 4, we tested the hypotheses on another sample of the data.

4. Research context and data

LEGO Ideas was launched in 2008 as an in-sourced continuation of a platform initially started by LEGO fans. On LEGO Ideas, users submit their ideas for new sets that eventually could be commercialized. These ideas represent user-generated innovations as they are new combinations of existing LEGO bricks. Ideas are manifested as project proposals,

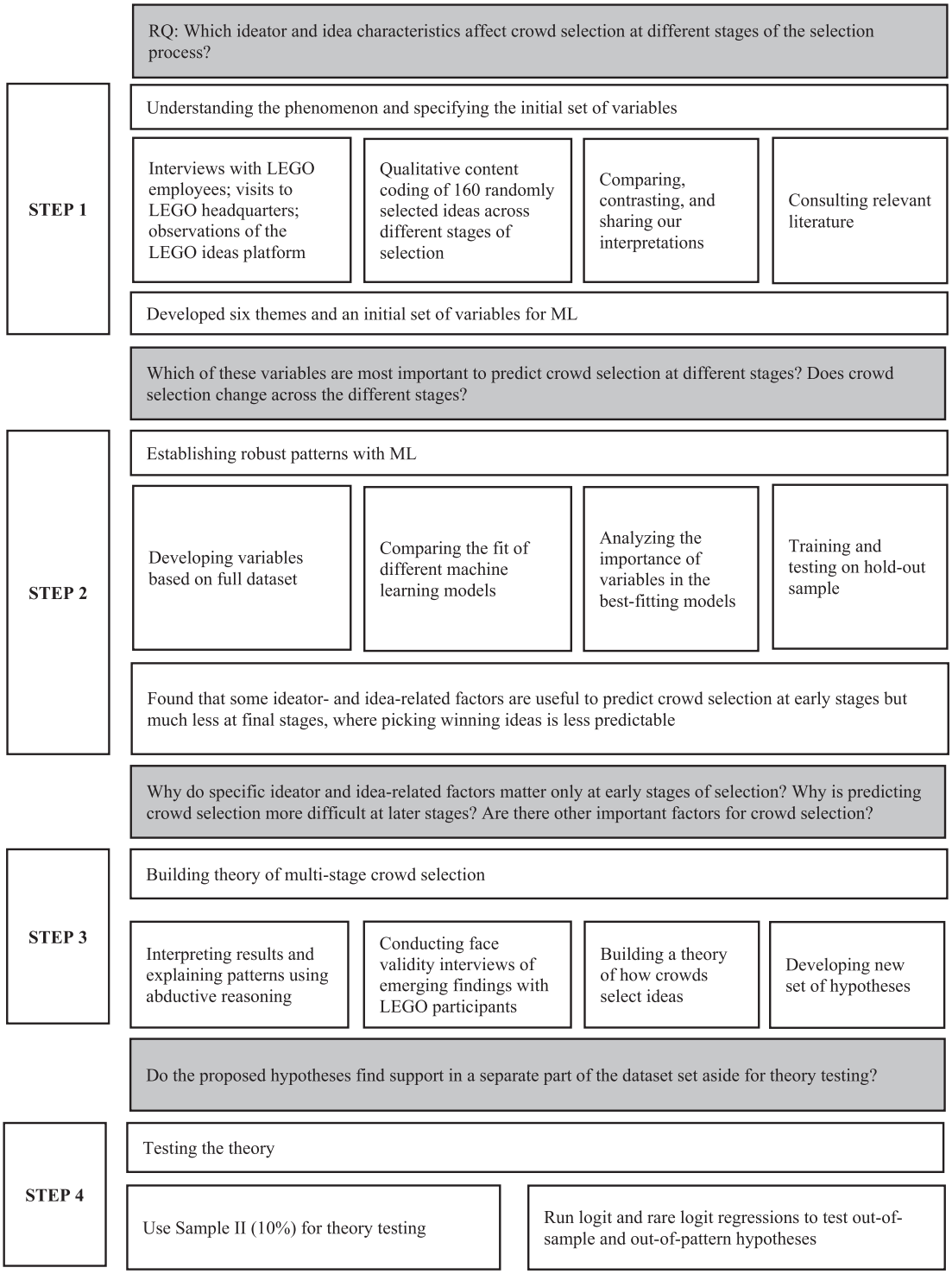


Fig. 1. Overview of research design and analytical steps.
(Adapted from He et al. (2020).)

which we refer to as ideas and projects. An idea on LEGO Ideas consists of a built model (i.e., a prototype), a written description, and images framed on neutral backgrounds. Ideas must follow a few guidelines, such as incorporating no more than 3000 LEGO bricks, violating no third-party or LEGO property rights, and containing no controversial content. These restrictions aside, almost any idea is welcome, and more than 80,000 user-generated tags show the diversity of ideas. After users publish a project on the platform, they can make limited modifications, updates, or refinements. LEGO Ideas is interactive and engaging; users comment on each other's projects and follow each other's activities.

The crowd selects ideas through a voting system, and ideas must go through multiple stages of selection to eventually become an official LEGO set. Users can vote only once for each project and cannot see which crowd members supported which idea. They only see an aggregate number of votes for a certain idea at a certain time, but not who voted. Voting for a specific project does not commit a user to purchase the product but induces a cognitive cost of choosing to support an idea—and being asked follow-up questions about the support. The crowd

selection process on LEGO Ideas has a multi-stage design with four stages, as shown in Appendix 1. New ideas have 60 days to obtain 100 votes. If this milestone is reached, the idea's lifetime on the platform is extended, granting it one year to reach 1000 votes. If 1000 votes are reached, additional six months are granted to reach 5000 votes, and if that goal is also met, six more months are granted to reach 10,000 votes. At each stage, the projects that do not meet the voting thresholds are dropped from the platform. If an idea gets 10,000 votes, it qualifies for review by LEGO's new product development (NPD) team, who decides whether the idea becomes an official product. The NPD team holds three rounds of reviews each year and evaluates projects based on originality, commercial potential, and fit with the brand. If the idea becomes an official product, the original idea is refined by LEGO for commercialization and marketed and distributed through its multiple channels. The ideator receives 1 % of the product's total net sales. Credits and a short ideator biography are printed on the products. Many user-developed LEGO sets have generated significant profit for both the ideators and the company.

LEGO shared all platform data with us in compliance with the European Union's GDPR data regulation (Tankard, 2016). Our data are unique as they include not only the currently active projects but all projects that failed and are thus no longer available on the platform. This is vital for our research question and circumvents success bias that can distort organizational research (Denrell and Kovács, 2008). LEGO only collects limited demographic data at the user level, but measures such as age and gender were available. At the same time, there is a wealth of data regarding the user's activities on the platform. These include logins, users followed, and comments (including their content). The data contain detailed descriptions, tags, and images for each idea. Since we are interested in understanding the importance of ideator and idea characteristics at different stages of crowd selection, we aggregated the data to the idea level. Our unit of analysis is the idea.

5. Findings

5.1. Step 1: understanding the phenomenon and specifying the initial set of variables

At the start of the study, we visited LEGO multiple times and held ten meetings to understand the context of the LEGO Ideas platform and its role within the company. We conducted five informal interviews with the community manager and other managers involved in the platform's management. These qualitative insights helped us understand the platform's history, current design, future plans, and how decisions are made. For example, we learned that LEGO organizes the crowd selection as a multi-stage process not only to arrive at a narrow set of promising ideas for its review. They also do so to make the platform more attractive. The community manager explained: "We introduced the milestone of 100 votes in 60 days; the main reason was to reduce the number of low-quality submissions. We wanted to filter out low-quality projects as quickly as possible because we found out that ideas that do not reach 100 supporters within 60 days… they will never get any better." The first milestone of reaching 100 votes in 60 days is thus the most crucial step because it populates the platform with ideas (increasing project lifetime from 60 days to 425 days). Between January 2016 and July 2018, 42,041 ideas with complete data about the ideator and idea were launched on LEGO Ideas. By September 2018 (60 days later), 3348 (7.96 %) reached 100 votes within the first 60 days.

Our goal in Step 2 is to identify factors that predict which ideas reach the first and subsequent milestones in multi-stage crowd selection. Therefore, we predicted which ideas got 100 votes (in 60 days), 1000 votes (in an additional 365 days), 5000 votes (in an additional 182 days), and 10,000 votes (in another 182 days). We defined the projects reaching each of these milestones as events of success in terms of providing content to the platform and the remaining projects as failures in this regard. We use the corresponding binary variables as dependent variables in our ML models.

To arrive at an initial set of variables that could explain crowd selection at different stages, in our informal interviews we asked the community manager and other employees involved in the management of LEGO Ideas which variables they thought had the most bearing on which ideas get selected. Informed by these insights, we conducted a content analysis and examined stratified random samples of ideas that had reached different levels of crowd support. We randomly sampled 40 ideas at each selection stage: up to 100, up to 1000, up to 5000, and up to 10,000 votes. Two authors carefully investigated each of the 160 ideas and derived a list of variables that likely affect crowd selection. We also analyzed the ideators' profiles to derive ideator-related characteristics. After doing this independently, we shared our interpretations and discussed whenever we disagreed. Finally, we consulted literatures on the evaluation of creative ideas (creativity), the funding of new ideas (entrepreneurship), and consumers' evaluations of product ideas (consumer behavior) to ensure that our variables are in line with prior work. We focused on identifying variables that were widely used to explain idea evaluation and selection, translate to the crowdsourcing context, and we could measure.

Based on this three-step procedure (interviews with LEGO, content analysis of ideas, literature review), we derived six major themes that explain why some ideas gain crowd support: ideator attributes, ideator experience, ideator status, idea topic, idea image characteristics, and idea text characteristics. Appendix 2 summarizes the six themes and the associated variables with their definitions. We operationalized the variables in line with common measures used in prior work. The initial set of variables is the starting point for the ML approach in Step 2. Given its importance, we detail the variable selection below and provide descriptive statistics in Table 1.

5.1.1. Ideator attributes

Conversations with LEGO, the qualitative analysis of the 160 ideas, and the literature suggested that ideator characteristics matter for crowd selection. Many users clicked on the ideator's profile to learn more about them. We learned that *Tenure* is essential, as ideators with greater tenure seemed to obtain more support for their ideas. It takes time for users to understand an online community's social structure and functioning (Hanson et al., 2019). Users with greater tenure have more experience with the platform and may thus be better equipped to propose ideas that gain support (Li et al., 2016). Moreover, creativity and entrepreneurship research highlight that individuals consider biographic information about the ideator when assessing ideas, especially age and gender. The relationship between age and creativity varies over time (Jones and Weinberg, 2011). In our qualitative investigation, we noticed that *Age* is important because LEGO Ideas attracts both kids and adults, and adults are financially better positioned to make new creations. Regarding gender, studies have found that people stereotypically associate men with more creative behaviors and that men's ideation efforts tend to receive more favorable evaluations than women's (Luksyte et al., 2018). In line with this, the comments on the 160 ideas suggested that the ideator's *Gender* mattered for crowd selection.

5.1.2. Ideator experience

Creativity research points to the importance of experience for creating high-quality ideas. This is based on the notion that good ideas come from new and original combinations of a person's accumulated knowledge and experience (Dahl and Moreau, 2002). In studying the ideators, we learned that they had accumulated different experiences through their activities on the platform. In line with prior research, we saw that *Projects launched* (Bayus, 2013) or *Previous success* (Deichmann and van den Ende, 2014) enabled ideators to learn about the ideation process, which can translate into proposing higher-quality ideas in the future (Jensen et al., 2014) and ultimately become more successful at gaining crowd support. Past success also helps ideators to establish a positive reputation as successful innovators, increasing the chances for

Table 1
Descriptive statistics and correlation matrix.

Variables		Mean	SD	Min	Max	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Outcomes														
1	Idea reaching 100 votes	0.09	0.28	0	1									
2	Idea reaching 1000 votes	0.01	0.12	0	1	0.38								
3	Idea reaching 5000 votes	0.00	0.06	0	1	0.18	0.48							
4	Idea reaching 10,000 votes	0.00	0.05	0	1	0.15	0.39	0.77						
Ideator attributes														
5	Tenure	7.73	12.98	0	87	0.23	0.11	0.05	0.05					
6	Age	27.07	13.82	12	117	0.12	0.06	0.03	0.02	0.15				
7	Gender	1.05	0.33	0	2.00	−0.02	0.00	−0.00	−0.00	−0.01	0.17			
Ideator experience														
8	Projects launched	6.92	17.20	0	200	0.10	0.03	0.00	−0.00	0.38	0.11	0.00		
9	Previous success	0.14	0.35	0	1	0.34	0.15	0.06	0.05	0.48	0.15	−0.02	0.47	
10	Logins	128.53	473.54	1	14,289	0.15	0.07	0.03	0.02	0.36	0.13	−0.01	0.67	0.45
11	Votes cast	98.25	412.19	0	11,290	0.09	0.02	0.01	0.00	0.29	0.10	0.00	0.60	0.36
12	Comments written	145.45	893.17	0	26,417	0.08	0.04	0.01	0.01	0.25	0.10	0.00	0.59	0.15
13	Users followed	66.45	253.40	0	4205	0.18	0.10	0.05	0.03	0.34	0.12	−0.00	0.66	0.45
14	Sentiment of comments	0.11	0.74	−16.25	9.09	0.08	0.02	0.01	0.01	0.06	0.05	0.00	0.10	0.15
15	Variation in sentiment	0.59	0.89	0	7.83	0.15	0.06	0.03	0.02	0.36	0.06	0.00	0.36	0.41
Ideator status														
16	Total votes received	706.56	3668.79	0	71,635	0.29	0.26	0.11	0.07	0.27	0.13	0.01	0.45	0.44
17	Highest number of votes received	224.36	1102.64	0	10,126	0.33	0.30	0.15	0.11	0.25	0.12	0.00	0.25	0.46
18	Personal followers	23.44	312.49	0	8326	0.13	0.17	0.06	0.02	0.04	0.04	−0.01	0.06	0.18
19	Unique followers	991.26	9326.67	0	368,152	0.07	0.02	0.00	−0.00	0.18	0.11	0.00	0.38	0.23
20	Comments received	184.44	844.98	0	21,103	0.16	0.10	0.04	0.03	0.32	0.13	−0.00	0.72	0.42
21	Sentiment of comments received	0.27	0.50	−2.44	7.54	0.11	0.03	0.01	0.00	0.13	0.02	0.02	0.20	0.28
22	Variation in sentiment received	0.57	0.88	0	8.34	0.21	0.08	0.03	0.02	0.40	0.06	0.02	0.43	0.55
Idea topics														
23	Number of tags	7.87	8.60	1	224	0.20	0.10	0.07	0.06	0.17	0.02	−0.01	0.14	0.26
Idea image characteristics														
24	Number of images	6.78	6.82	1	151	0.17	0.09	0.05	0.04	0.12	0.04	−0.00	0.08	0.16
25	Size of the principal image	866.74	877.70	0	7699	0.00	0.02	0.01	0.01	−0.02	0.02	0.03	−0.08	−0.03
26	Colorfulness	40.76	24.37	0	185	−0.01	−0.00	0.00	0.01	0.03	0.03	0.03	−0.00	0.00
27	Complexity	0.00	0.00	0	0.02	0.05	0.04	0.02	0.02	0.04	−0.02	−0.00	0.02	0.07
Idea text characteristics														
28	Length of description	779.98	830.90	4	8008	0.30	0.17	0.10	0.08	0.26	0.10	−0.00	0.19	0.31
29	Sentiment of description	2.99	6.12	−37.15	65.77	0.20	0.12	0.07	0.06	0.12	0.09	0.02	0.06	0.16
30	Resubmission count	0.78	0.80	0	8	−0.16	−0.07	−0.04	−0.03	−0.10	−0.04	0.00	−0.17	−0.16

future ideas (Bayus, 2013). We also found that ideators accumulated experience in other ways, which are studied less by prior work. Some users simply gained experience by spending time on the platform, as indicated by frequent *Logins*. The interviews showed that engaging with other users and their ideas are opportunities to learn indirectly. Supporting this, Riedl and Seidel (2018) found that indirect learning experiences can help ideators to develop better ideas. Ideators actively engaging in a wide range of activities on the platform are better equipped to identify emerging needs and trends within the community and can incorporate them into their ideation efforts (Jensen et al., 2014). We thus included *Votes cast*, *Comments written*, and *Users followed* to capture experience and indirect learning. Informal conversations with the community manager revealed the importance of these activities: "Remaining active is crucial for getting support on future projects. These users work on maintaining engagement by being active in the community, especially by supporting and commenting on other projects and by submitting new projects. This helps them get attention."

Beyond the sheer number of interactions, their content also matters (Perry-Smith and Mannucci, 2017). Some users accumulated experience by leaving detailed comments on others' ideas. Being positive—indeed, "Everything is Awesome," as the LEGO movie says—was important to establish social relationships with others. For example, a user maintaining a positive tone in the *Sentiment of comments* is perceived as positive and encouraging (Jensen et al., 2014). In contrast, comments with a negative tone can be interpreted as critical or even hostile (Füller et al., 2014). However, only "spreading the love" may not be seen as authentic either. We discovered that some users showed more *Variation in sentiment* across comments than others. If the variation is large, the user may be an eccentric ideator valuing creativity and honesty more than social conformity (Garimella et al., 2018). We noticed that other users perceived such ideators as critical and honest—for instance, one ideator replied to a comment on his idea: "It means a lot coming from you, as I know you have high standards."

5.1.3. Ideator status

When evaluating ideas, individuals tend to consider an ideator's status. Status signals competence and credibility (Zhou et al., 2019) and indicates the position in the community built through repeated social interactions with peers (Chen et al., 2020). We noticed that some ideators accumulated status by engaging in various activities on the platform. This was echoed in interviews with LEGO: "We see that once people get the first 1000 supporters, they start building a name in the community; they create a brand for themselves that helps them attract more attention to new projects. They are very likely to gather support for their new projects, and they would have an easier time in getting 10,000 supporters due to the name they built in the community."

Studying the 160 ideas and their ideators, we discovered multiple facets of status. In broad terms, ideator status can manifest in the support their projects obtained, the followers they attracted, and the comments they received (Füller et al., 2014). In line with the quote above, the *Total votes received* appeared to be correlated with future ideas being selected. Ideators learn from experience and are motivated by their success (Deichmann and van den Ende, 2014), so we measured the *Highest number of votes received* as a signal of prior success. A previously highly popular idea strongly signals the ideator's ability. Similarly, some ideators were able to attract a significant number of *Personal followers* and

(3)	(10)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)	(29)
0.77																	
0.05																	
0.03																	
−0.00																	
0.00																	
0.06																	
0.03																	
0.01	0.77																
0.01	0.77																
0.05	0.71																
0.01	0.11																
0.03	0.28	0.15															
0.11	0.42	0.07	0.27														
0.15	0.29	0.07	0.29	0.84													
0.06	0.12	0.09	0.12	0.38	0.50												
0.00	0.57	0.04	0.12	0.30	0.17	0.01											
0.04	0.75	0.08	0.22	0.60	0.40	0.61	0.10										
0.01	0.11	0.26	0.35	0.10	0.12	0.03	0.08	0.06									
0.03	0.31	0.14	0.64	0.26	0.29	0.12	0.12	0.27	0.65								
0.07	0.12	0.09	0.30	0.15	0.15	0.05	0.08	0.12	0.20	0.30							
0.05	0.06	0.09	0.21	0.08	0.09	0.06	0.01	0.30	0.17	0.21	0.03						
0.01	−0.05	0.00	−0.10	−0.03	−0.01	0.01	−0.03	−0.03	−0.06	−0.10	0.05	−0.05					
0.00	0.00	−0.01	−0.01	−0.00	0.01	−0.02	−0.01	0.00	−0.02	−0.01	−0.02	0.04	−0.01				
0.02	0.03	−0.00	0.05	0.06	0.07	0.01	0.02	0.05	0.01	0.05	−0.00	0.12	0.22	0.03			
0.10	0.18	0.10	0.31	0.20	0.22	0.07	0.07	0.42	0.17	0.31	0.38	−0.03	0.02	0.07	0.17		
0.07	0.07	0.12	0.11	0.10	0.12	0.06	0.02	0.22	0.09	0.11	0.19	−0.01	−0.01	0.03	0.29	0.06	
−0.04	−0.13	−0.06	−0.14	−0.10	−0.09	−0.07	−0.04	−0.12	−0.14	−0.19	−0.08	−0.04	0.04	0.00	−0.02	−0.09	−0.07

Unique followers who can help garner support for ideas (Sun and Rui, 2017). Many followers hint at the ability to create ideas with so much charm that other users pro-actively sign up to stay tuned to such ideas (Sun and Rui, 2017). Finally, the number of *Comments received* shows the ideator's prestige (Higgins and Gulati, 2003). It captures the ideator's ability to be recognized for generating creative ideas, capturing attention, and arousing curiosity (Füller et al., 2014). In reviewing the comments on the 160 ideas, we learned that the *Sentiment of comments received* matters. Some ideators mainly attract positive comments, whereas others receive more neutral or negative comments. Positive comments show that the idea has attracted endorsement from other users (Perry-Smith and Mannucci, 2017), which may signal higher quality (Jensen et al., 2014; Courtney et al., 2017). At the same time, some ideas spark more controversy than others (Garimella et al., 2018), which can help to attract attention and support. Such controversy is reflected in the *Variation in sentiment received* and can also be interpreted as a measure of creativity (Garimella et al., 2018).

5.1.4. Idea topics

The topic of an idea is an essential determinant of crowd selection, as specific topics may appeal more to crowd members and attract greater support. Supporting this notion, Mollick (2014) has shown that crowd support varies systematically across product categories. In coding the 160 ideas, we found differences in their topics, ranging from buildings, fantasy, and heroes to nature and vehicles, to give a few examples. The different idea topics seemed to affect the likelihood of crowd selection, and one community manager explained their importance: "The theme of the model is an important factor. For instance, anything that links back to IP has a much broader appeal to people in and outside the community. For instance, ideas linked to TV shows are likely to be very successful and of high quality as well. These ideas have a sizeable community behind [them] that is very interested in the theme and would like the idea to become a LEGO set." We derived 15 latent *Idea topics* through topic modeling via the Latent Dirichlet Allocation (LDA) (Blei et al., 2003) (for details, see Appendix 2). In addition, the *Number of tags* captures the ideator's strategy to classify the project and link it to certain (popular) topics.

5.1.5. Idea image characteristics

When evaluating ideas, users tend to rank ideas that are rich in appearance higher (Franke et al., 2009). One way to create such an appealing appearance is by using images conveying high levels of detail (Li et al., 2016). On LEGO Ideas, the ideas manifest themselves as spatial constructions of colorful bricks, and the best ideas seem to have high visual appeal. Although the proverb suggests that "a picture is worth a thousand words," organization and innovation research rarely study images as explanations for (crowd) selection. Our qualitative coding suggested that images are as important as text descriptions. This is echoed by one LEGO manager: "The images have to be very clear and detailed, and they need to show the creation from multiple angles. This is important for people to get a full understanding of what the model is about." The principal image is essential as it appears as the thumbnail on the LEGO Ideas main page. Interviews with managers highlighted the importance of the principal image for crowd selection: "The thumbnail image is going to appear on the main page of the website, so it is highly critical that it conveys well the content and details of a project. So, the attention and subsequent idea support are likely to already be brought by the thumbnail image." Our qualitative coding revealed important differences in the *Number of images* and the *Size of the principal image*. Using more images can make the idea more concrete and

understandable (Li et al., 2016; Courtney et al., 2017). The size of the principal image is particularly important for a favorable early impression, encouraging users to explore the idea further (Lam et al., 2007). Moreover, the *Colorfulness* and *Complexity* of the images varied. Diversity of color usage and image complexity capture the salience of an image, and salience attracts attention (Hays et al., 1972). To capture such differences and do justice to the images' importance, we extracted these features from the images, as described in Appendix 2.

5.1.6. Idea text characteristics

Besides images, the text describing an idea is a central part of the presentation that varied among the ideas we analyzed qualitatively. One platform manager explained the importance of the idea description: "The description needs to describe the project thoroughly for people to understand it. These are basic elements, foundational elements for a good idea to get support." The *Length of description* captures the ideator's cognitive investment to offer a thoughtful and informative idea description (Beretta, 2019). Positivity is contagious, and an enthusiastic tone in the idea description (*Sentiment of description*) may spark empathy and excitement among other users (Kramer et al., 2014; Parhankangas and Renko, 2017). Among the reviewed ideas receiving little support, many had a high *Resubmission count*. Some projects do not comply with the formal requirements in their first submission and therefore get rejected. Projects that start poorly with one or more rejection(s) seem less likely to be selected, even when they comply with the requirements, and are later granted access to the platform. One reason could be that these ideas may now comply with the official rules but may not fully comply with social norms and therefore receive less favorable evaluations (Gatzweiler et al., 2017).

5.2. Step 2: establishing robust patterns with ML

We follow the suggestions by Shrestha et al. (2021) for building theories using ML. One central function of ML is feature selection, that is, to identify important predictors for a specific outcome (He et al., 2020). In line with this, we use ML models to identify the factors that drive crowd selection at different stages of LEGO Ideas' multi-stage contest. We start with the initial set of variables identified in Step 1, then employ ML to find the most important predictors and prune the possible explanations.[4] Feature selection removes redundant variables and prevents overfitting caused by adding variables with little predictive power (see also He et al., 2020).[5]

One advantage of using ML for feature selection is that it performs better than simple linear models (e.g., logistic regression) when the data do not follow a linear pattern, especially when data are imbalanced—that is when only a few data points belong to the minority class (Davis and Goadrich, 2006). In our case, the minority class is successful ideas, and the data are highly imbalanced, with less than 1 % successful ideas at the final stage of crowd selection. A linear regression model hardly detects a pattern in such a situation. Another advantage of ML methods is that they are tunable based on specific performance measures (Abu-Mostafa et al., 2012). This allows us to find the most effective features for predicting crowd selection instead of just improving the model's accuracy. This is important with imbalanced data because models focusing on accuracy tend to be biased toward the majority class (in our case, unsuccessful ideas). The interpretability of ML models decreases as they become more complex and powerful in capturing nonlinearities. Considering this complexity-interpretability trade-off, we used both linear (easy to explain) and nonlinear (hard to explain) ML models.

While we primarily use ML algorithms that reveal variable importance, some models also show the direction of effects.[6] As a baseline, we used logistic regressions with all independent variables (Friedman et al., 2001). Following He et al. (2020), we chose the Least Absolute Shrinkage and Selection Operator (LASSO) implemented as a binomial (successful or unsuccessful) linear probability model (Tibshirani, 1996; Zhao and Yu, 2006). LASSO is a regression method selecting variables via regularization. Regularization shrinks the coefficient estimates toward zero, which makes the model less prone to overfitting as it discourages learning a too complex model. The variable selection also makes the model more parsimonious. Therefore, we expect LASSO to produce better predictions with simpler models than complete regression models. One of the greatest advantages of LASSO is its interpretability. To preserve this feature, we did not add any higher-order polynomials or interaction terms (Shrestha et al., 2021). Instead, we complemented LASSO with a Gradient Boosting (GB) algorithm as a robustness check (Breiman, 2001). The benefit of GB is that it captures nonlinearity among features, and we can compare its performance with LASSO.[7]

5.2.1. Data preparation

Our level of analysis is the idea because our goal is to predict the success of ideas at different stages of crowd selection. Following Shrestha et al. (2021), we employ a two-step method and split our data into two samples: a sample for pattern detection (Sample I used in Step 2) and a sample for theory testing (Sample II used in Step 4). Observations were randomly assigned to one of the samples. As our data is highly imbalanced at all stages of selection, we assigned 90 % of the observations to Sample I for pattern detection (Shrestha et al., 2021). This increases our chance of learning because it gives us more data points in the minority class of successful ideas. We left 10 % of the sample untouched, assigned it to Sample II, and used it for theory testing in Step 4. Within Sample I, we used 80 % of the observations to train our algorithms (training sample) and the remaining 20 % to tune the models with the unseen data (validation sample) (Shrestha et al., 2021). We used the Synthetic Minority Oversampling Technique (SMOTE) on the training sample to generate new (synthetic) data points for the minority class based on existing data. This makes the training dataset more balanced and increases the performance of the algorithm (see Appendix 3). Overall, Sample I contained 30,272 ideas in the training sample plus 7568 ideas in the validation sample (37,840 ideas in total), and Sample II contained 4203 ideas. The proportion of early-stage success (reaching 100 votes in 60 days) among all data is 7.96 % (3348/42,041), 7.96 % for Sample I (2410/30,272 in the training sample, and 603/7568 in the validation sample), and 7.97 % for Sample II (335/4205).

Prediction models assume a stationary process, where variable importance and effects are constant over time (He et al., 2020). We use data from three years (2016–2018) because language use in online platforms is dynamic (Danescu-Niculescu-Mizil et al., 2013). To check for stationarity, we ran an augmented Dickey-Fuller test, which showed that our data were stationary (see Appendix 4). Finally, to make the

[4] We also add additional controls for *Day of week* and the *Month*, which may influence the likelihood of ideas gaining support (Hartley and Cabanac, 2017).

[5] Overfitting is a modeling error that occurs when a function is too closely fit to a limited set of data points and therefore not flexible enough to come close to the shape of new unseen data points.

[6] Almost all ML packages can generate a list of important features. However, some drop the features with zero effect (or coefficient). Even if we have a list of ordered important features in some ML models, their interpretation is difficult because the order does not show the complexity of the patterns in terms of interaction or higher-order terms. Therefore, we consider both linear and nonlinear algorithms to compare their performance and opt for the linear ones if the others cannot improve the performance measure significantly.

[7] In supplementary work (not reported here due to space), we compared our models with random forests and decision-trees LASSO. In terms of model performance (measured through the AUPRC, see below), LASSO and GB were among the best-performing algorithms. We used these algorithms because they also perform feature selection, and they do so in different ways.

coefficients of the different interval-scaled variables comparable, we normalized all variables to values between 0 and 1.

5.2.2. Model performance

An integral part of ML is to find the best-fitting model. To assess model performance, we used criteria designed explicitly for binary classifications. ML algorithms rarely separate two groups perfectly (e.g., as "successes" and "failures"). Instead, the predicted probabilities often overlap. For every probability threshold translating a success probability between zero and one into a prediction of either zero (i.e., failure) or one (i.e., success), there are true positives and true negatives but also false positives and false negatives. We used precision and recall measures to assess model performance, which are useful when the dataset is highly imbalanced (Davis and Goadrich, 2006). Precision is defined as True Positives divided by (True Positives + False Positives) or the proportion of all successful ideas detected by the algorithm that are true successes. Recall is defined as True Positives divided by (True Positives + False Negatives) or the proportion of all successful ideas detected by the algorithm. These measures capture how good the algorithm is in not identifying failures as successes (precision) and in not missing successes (recall). As our focus is on "not missing successful ideas," we used recall as our primary measure.[8]

Comparing the ML models with random guessing as the baseline identifies their power. A two-sample *t*-test for proportions showed that LASSO and GB improved random guessing significantly (Appendix 5). Similarly, the McNemar's test showed that the difference was statistically significant (Appendix 5) and practically significant: random guessing had a recall of 7.96 %, so the ML models were roughly ten times better (81.09/7.96). Another important benchmark is full logistic regression (Shrestha et al., 2021). While full logistic regression performed well in isolation (recall 48.42 %), it was outperformed by the ML models. Compared to the best-performing algorithm, the GB, which had a recall of 81.09 %, there was a difference in recall of 32.67 %, and thus the relative improvement of using ML was 410 % ((81.09–48.42)/7.96). Although the full logistic regression provided a reasonably good forecast, ML algorithms significantly improved the prediction while performing feature selection, thus providing a more parsimonious model. Given that both LASSO and GB performed well on hold-out validation data and McNemar's test showed no clear difference between the two, we focused on LASSO because of its better interpretability and used GB as a robustness check.

Fig. 2a through d shows the Precision-Recall (PR) curves plotting all ideas from left to right, beginning with the idea with the highest predicted probability of success, continuing with the second highest, and so forth. The PR curve plots the precision rate as a function of the recall (Davis and Goadrich, 2006). Each probability was translated into a binary prediction (success or failure) and was then compared with the actual outcome. Each point on the PR curve represents a precision-recall pair and corresponds to a selection threshold. The area under the precision-recall curve (AUPRC) ranges between the baseline of random guessing (here 7.96 %) and one, implying predicting all successes correctly. An AUPRC of 0.8109 would imply an 81.09 % probability that the model correctly classifies a "success."

5.2.3. Results

Table 2 summarizes the results from the LASSO, showing the variables identified as important in the feature selection based on the recall measure. To arrive at the predictions, we ran the model and tuned the parameters based on cross-validation (Abu-Mostafa et al., 2012) (see Appendix 7). As we normalized the variables, the coefficients identified each feature's importance and direction for crowd selection at the different stages. As a robustness check, we also used GB (for details and results, see Appendix 8) and performed feature selection based on the F1 measure (for results, see Appendix 9).

The results offer two central insights. First, the models explain crowd selection at early stages but much less at later stages. The algorithm identified 12 factors that helped predict crowd selection for the first stage (100 votes) but only 4 or 5 for the later stages. This was also reflected in Fig. 2a to d, where the AUPRC became smaller at each selection stage, indicating the decreasing ability of the algorithms to classify the ideas correctly. Second, the results shed light on *which* factors drive crowd selection at the different stages. Most notably, *Previous success* predicted crowd selection at all stages. A few factors helped to predict crowd selection at more than one stage: *Tenure* (100 and 10,000 votes), *Number of tags* (100 and 10,000 votes), *Variation in sentiment* (1000 and 10,000 votes), *Colorfulness* (100 and 1000 votes), *Length of description* (100 and 5000 votes), and *Sentiment of description* (100 and 5000 votes). Ultimately, however, our models were most useful in explaining crowd selection at early stages. Therefore, we focus on the factors predicting crowd selection at the first stage.

Many of our results at this stage are in line with prior findings from crowd selection: At the ideator level, we found that attributes such as *Tenure*, *Age*, and *Gender* matter. Ideator's *Previous success* and status (in terms of *Total votes received*) also helped to predict early crowd selection. At the idea level, the idea topics reflected in the *Number of tags* played an important role, just as image characteristics (*Number of images* and *Colorfulness*) and text characteristics (*Length of description* and *Sentiment of description*). These results largely confirm prior literature, as reviewed in Step 1. This also holds for the result that ideas with a high *Resubmission count* were less likely to succeed. A high number of resubmissions captures not only experience (which should have a positive effect on crowd selection) but also careless, low-quality idea presentations, especially given we captured experience through other variables.

While multiple factors helped to explain crowd selection, it is also interesting to consider what we *did not* find: several explanations from prior work have negligible or zero effects in our multi-stage contest (see Choudhury et al., 2021). For instance, the prestige of an ideator (in terms of the number of *Comments received*). Previous success and high support levels trump the number of comments received as a status signal at early stages of crowd selection. In contrast to suggestions by Riedl and Seidel (2018), indirect learning experiences gained from interacting with others' ideas (in terms of *Votes cast*, *Comments written*, and *Users followed)* were not an important predictor. While crowds' positive feedback (in terms of *Sentiments of comments received*) is argued to affect idea selection (Jensen et al., 2014), we found such feedback less relevant in multi-stage contests. The null findings illustrate the strength of ML, as it helps us prune theoretical explanations and reduces the risk of focusing on factors with little explanatory power.

Overall, many of our ML results aligned with prior work on crowd selection. At the same time, the ML feature selection helped us to prune factors that prior work has suggested but have little explanatory power. Beyond taking stock of different explanations from single-stage crowd selection and whether they hold in multi-stage selection, the ML approach (Step 2) and the qualitative approach (Step 1) also detected new patterns and brought forward new aspects of multi-stage crowd selection. These insights enable us to develop new theory in Step 3.

5.3. Step 3: building theory of multi-stage crowd selection

To build theory on multi-stage crowd selection, we focus on novel aspects of the empirical patterns detected in Steps 1 and 2. As ML algorithms reveal data patterns that are difficult to observe with other methods, it is important to assess and explain these patterns theoretically (He et al., 2020). We reviewed prior literature and conducted additional interviews in 2020, two with LEGO managers and eight with select ideators (with different experiences and outcomes) to gather their

[8] We also included the results of improving the F1 measure (both "not missing successful ideas" and "not considering unsuccessful ideas as successful") as a robustness check. More explanations about the performance measures in binary classification problems are reported in Appendix 6.

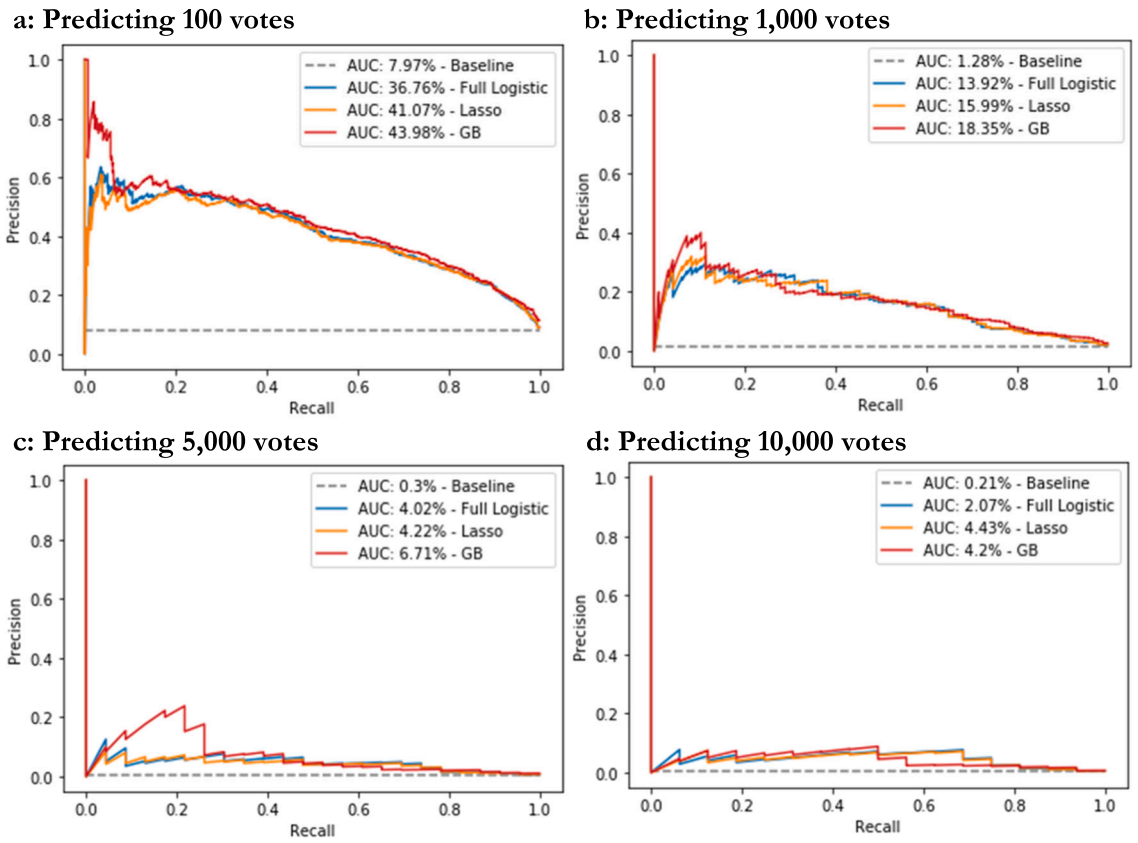


Fig. 2. a: Predicting 100 votes.
b: Predicting 1000 votes.
c: Predicting 5000 votes.
d: Predicting 10,000 votes.
Notes: The Precision-Recall curves are calculated as follows: Precision = True Positives / (True Positives + False Positives). The perfect model does not have any False Positives, so Precision would be 1. It means we have a horizontal line at the top of the chart (Precision = 1). Recall = True Positives / (True Positives + False Negatives). Recall identifies how many positives our model has not detected. If our model does not miss any positives, then the perfect model would be a vertical line at the right of the chart (Recall = 1). The gray line is a baseline random guess. Our models outperform random guessing for all four outcomes.

views on our findings and ensure face validity. These interviews were transcribed and coded to identify recurring themes based on detected similarities. During the coding process, we iterated between data and prior literature. These interviews helped us understand differences in factors driving crowd selection in early vs. later stages, pointing to the importance of social influence, promotion efforts, and fast traction in later stages Based on the insights from the interviews and prior literature, we developed two main hypotheses about the differences between the early vs. late stages of crowd selection (H1) and fast traction (H2). H1 is out-of-sample, focusing on explaining the patterns identified in Step 2. H2 is out-of-pattern, focusing on a new association that was not checked or observed before but emerged during our abductive theorizing and interviews (Shrestha et al., 2021).

5.3.1. *Early vs. late stages of crowd selection*

A central finding from Step 2 is that we can predict crowd selection at early stages but much less at later stages. Certain ideator and idea characteristics matter for predicting which ideas are weeded out in the early stages of crowd selection. However, these factors are less useful for predicting winners at later stages of crowd selection. We build on signaling theory (Spence, 1973) to explain our main finding.

Signaling theory suggests that individuals can resolve information asymmetries and uncertainty by communicating—or signaling—latent or unobservable quality and intent through their actions (Stiglitz, 2000). In crowdsourcing, ideators typically know more about their ideas than the other crowd members, leaving them uncertain about the idea's quality and potential. In this situation, certain ideator and idea characteristics can serve as signals for crowd members to infer latent idea quality. To be effective, signals need to be observable and costly (Connelly et al., 2011). The cost to acquire a signal is critical to signaling theory because it determines who obtains the signal and how it is interpreted (Spence, 1973). Receivers prioritize costly signals, which weakens the effect of low-cost signals (Connelly et al., 2011). In line with this, we found that ideator's status and previous success help predict early crowd selection, presumably because they are costly signals that can only be achieved by able ideators making a serious and sustained effort. Similarly, a carefully crafted idea presentation with an elaborated description and many images requires time and effort, signaling the ideator's commitment to an idea. In contrast, other factors related to low-cost activities, such as commenting, voting, or following, have little predictive power.

Signaling theory can explain why certain factors are more predictive than others and why these factors mostly matter in the early stages of crowd selection. In dynamic environments like crowdsourcing, the information available for signaling constantly changes (Connelly et al., 2011). As new information emerges, the effect of signals can fade, and new signals may replace earlier signals, exerting a substitution effect (Xia et al., 2016; Song et al., 2020). This is especially true for multi-stage crowd selection. At the early stages, idea and ideator characteristics are the only information available to the crowd. As ideas progress to later stages, however, new signals emerge. We argue that these new signals replace the idea- and ideator-related factors, making them less useful for

Table 2
ML models performing feature selection at multiple stages of crowd selection.

Variables	Predicting 100 votes	Predicting 1000 votes	Predicting 5000 votes	Predicting 10,000 votes
	LASSO	LASSO	LASSO	LASSO
Ideator attributes				
Tenure	1.66			2.42
Age	2.27			
Gender	−1.13			
Ideator experience				
Projects launched		−7.90		
Previous success	1.68	2.38	1.16	1.90
Logins				
Votes cast				
Comments written				
Users followed				
Sentiment of comments	1.11			
Variation in sentiment				
Ideator status				
Total votes received	1.78			
Highest number of votes received			6.84	
Personal followers				
Unique followers				
Comments received				
Sentiment of comments received				
Variation in sentiment received		−4.74		−7.83
Idea topic				
—building				
—city				
—color				
—details				
—fame				
—fantasy				
—figure				
—game				
—hero				
—nature				
—space				
—vehicle				
—vessel				
—war				
—world				
Number of tags	10.04			23.65
Idea image characteristics				
Number of images	4.75			
Size of the principal image				1.59
Colorfulness	−0.98	−0.76		
Complexity				
Idea text characteristics				
Length of description	6.23		9.69	
Sentiment of description	5.61		8.29	
Resubmission count	−5.67			
Trends				
Day of the week dummies				
Month of the year dummies				
Observations				
Observations for training	30,272	30,272	30,272	30,272
Observations for testing	7568	7568	7568	7568
Model fit				
Baseline (random guessing) (%)	7.96	1.28	0.29	0.20
Precision (%)	30.39	7.78	2.75	2.43
Recall (%)	76.95	74.23	78.26	81.25
F1 (%)	43.57	14.09	5.32	4.73
MCC (%)	41.62	21.57	13.68	13.34
True positives	464	72	18	13
True negatives	5902	6618	6909	7031
False positives	1063	853	636	521
False negatives	139	25	5	3

predicting later-stage winners.

Passing earlier stages of crowd selection is a strong and informative signal. This was also highlighted by our interviewees, who pointed to the high standards of LEGO Ideas and explained that ideas passing the early selection stages are known to have a well-developed presentation or to come from highly recognized ideators. So, as ideas progress through the selection stages, their popularity becomes apparent to crowd members, who may interpret this social information as a signal of idea quality. Similarly, studies highlight that individuals in high-noise environments tend to pay close attention to endorsement signals provided by third parties (Connelly et al., 2011; Courtney et al., 2017). As these signals of social approval emerge, they replace earlier (idea- and ideator-related) signals. The further an idea advances, the more it is exposed to social influence processes, which can shape perceptions of quality and lead to unpredictable outcomes (Salganik et al., 2006). Individuals adjust their opinions to the opinions of others so that they converge toward uniform social behavior (Duan et al., 2009). This consensus formation is subject to social influences (Cattani et al., 2008), as one interviewee explained: "I think it is also a matter of having the right people picking up your idea at the right time. That makes a difference in whether you will be able to get to the 10,000 votes."

At the same time, passing early stages of crowd selection is not only a signal to the crowd. The ideator also receives new (positive) information about the idea's popularity, which can make them more invested in the idea. As a result, they may promote their ideas further, not only on the platform but also in their social network (family and friends) and external online communities (Davies and Giovannetti, 2018). Individuals who get to know the idea may then turn into additional supporters. One interviewee explained his promotion efforts in response to losing momentum after passing the early selection stages: "The idea reached 3000 votes in a relatively short amount of time, then it stagnated for a while. So, I had to do something to boost again the attention toward my project. I started to promote myself on social media channels as well as in local stores here where I live. I put a lot of effort and money into promoting my idea. I did advertising on Facebook, I have friends who know how to do campaigning on Facebook to get exposure. I became a member of 30 Facebook groups about LEGO fans." The quote illustrates that ideators promote their ideas to reach more and new crowd members at later stages and that much of the promotion occurs outside of LEGO Ideas. With increasing promotion efforts, the early idea and ideator-related signals lose importance and fade. In line with this, some interviewees mentioned that promotion efforts could play an important signaling function at later stages, with crowds paying more attention to this new signal.

Finally, a certain amount of randomness and luck is involved in reaching later stages. For example, Xin, the ideator from our vignette, stressed that her timing was lucky: "The other side was the timing of the way it gathered support because those were factors that are out of my control. I could only do the best I could in creating a nice product idea." For all these reasons, we hypothesize:

H1. In multi-stage crowd selection, idea- and ideator-related factors are important for crowd selection at early stages but less important for the final stage of crowd selection.

5.3.2. Fast traction

Compared to single-stage contests, multi-stage crowd selection is typically longer and more dynamic. Throughout the selection stages, ideas may gain or lose momentum. In line with this, our abductive approach helped us to identify another temporal factor that prior work has overlooked, namely the importance of gaining fast traction. Observations from the platform and our interviews suggested that fast support foreshadows crowd selection at later stages. Ideas that receive support quickly and reach many votes early are more likely to progress and reach the later stages of crowd selection. As stated by one interviewee: "It is very important for a project to get to 10,000 votes as fast as possible; if it starts stagnating, it can become difficult to maintain the interest of the community. From my experience, I noticed that projects that get votes faster at the beginning will be more likely to make it." Supporting this notion, a cluster analysis showed that the ideas can be grouped into two clusters and that these clusters differ visibly in the average number of votes received in the first weeks on the platform (see Appendix 10). The figures show a strong time dependence of crowd support, where early support foreshadows future support.

There are several reasons why fast traction facilitates crowd selection at later stages. One major reason links back to signaling theory; quickly attracting many votes may be perceived as a substantive signal of idea quality by the crowd. When crowd members see many people supporting an idea early, they believe it is good, which boosts support (Connelly et al., 2011). Fast crowd support can also trigger users to follow the decisions made by other users before them and thus can contribute to trigger social influence processes (Bikhchandani et al., 1998). In other words, fast traction is a very early signal that can emerge before passing the first stages of crowd selection. This early signal can be reinforced by platform architecture because trending ideas are often displayed at the top of the webpage, allowing them to garner further support. The algorithm governing LEGO Ideas rewards ideas gaining fast traction by increasing their visibility and displaying them prominently on the first webpage. It is also likely that fast traction generates word-of-mouth, as early supporters diffuse information about the idea to other crowd members (Colombo et al., 2015) and through external channels. A LEGO manager highlighted this process: "We see ideas that explode in terms of support because the community picks them up itself. Many supporters would even share the idea in their social media or other communities to attract more supporters to get to the 10,000 votes." In line with this, a user explained: "Votes coming quickly early on helped create a boost in support. Many people liked my idea and shared it too on social media. Getting the word out there, talking about it matters a lot. I was able to get to 10,000 votes in only 36 days." Gaining fast traction may not only foster word-of-mouth by other crowd members, as highlighted by the previous quotes, but may further motivate ideators to invest efforts in promoting their ideas to sustain the momentum and progress to the final stages of selection. These insights lead us to the following hypothesis:

H2. In multi-stage crowd selection, ideas gaining fast traction at early stages are more likely to progress to the final stage of crowd selection.

5.4. Step 4: testing the theory

In the previous steps, we engaged in an abductive process (Bamberger, 2018) of phenomenon-based theorizing (von Krogh, 2018). In Step 3, we generated new theoretical insights to explain our empirical patterns (He et al., 2020; Shrestha et al., 2021). We developed hypotheses by iterating between data and theory, which we test in this final step. It is important to test the out-of-sample and out-of-pattern hypotheses on a separate sample of the same data (Shrestha et al., 2021). Following Shrestha et al. (2021), we tested the hypotheses in Sample II, the 10 % of the data we did not use for machine learning and set aside for theory testing.

5.4.1. Early vs. late stages of crowd selection

To test H1, we performed a series of logit regressions. We selected a set of variables that our ML algorithms in Step 2 identified as important to predict crowd selection: *Projects launched*, *Previous success*, *Total votes received*, *Number of tags*, *Number of images*, *Length of description*, and *Sentiment of description*. We linked these variables to crowd selection at the first stage (100 votes; Model 1 in Appendix 11) and the final stage (10,000 votes; Model 2). In Model 2, we ran a rare event logit (Firth, 1993) to account for the small number of ideas reaching the final stage of crowd selection. An ordinary logit model would return biased estimates because of the small number of cases in the less frequent of the two categories (Firth, 1993). Since some ideators submitted more than one idea, observations are not independent, and we accounted for this by calculating robust standard errors clustered by the ideator.

Our models provided partial support for H1. Model 1 shows that all variables in the model had a statistically significant effect on crowd selection at the first stage. In line with our theorizing, the idea- and ideator-related factors are thus important for crowd selection at early stages. Our argument, however, is that the same factors are less important at the final stages. Model 2 shows partial support for this argument. The effects of two ideator- and idea-related factors remained statistically significant (*Total votes received*, *Number of tags*), and two more marginally statistically significant (*Number of images*, *Length of description*).

5.4.2. Fast traction

To test H2, we created a new variable, *Fast traction*, that measured the days an idea took to reach 100 votes. We ran another rare event logit regression and included this variable in our model to test whether ideas that gain fast traction are more likely to reach crowd selection at the final stage (Model 3 in Appendix 11). The number of observations used in this model was lower because only ideas that got 100 votes were included. The results show a statistically significant negative effect for *Fast traction* ($b = -2.636$; $p < .001$), which is in line with our theorizing and remarkable given that most of the other ideator- and idea-related factors were no longer significant in Model 3 (except for *Number of tags*). The longer it took for an idea to gain traction in the early stages, the less likely crowds were to select it at the final stage. We thus found support for H2.

6. Discussion

Much open innovation research focuses on how crowds *generate* ideas (Afuah and Tucci, 2012; Bayus, 2013; Jeppesen and Lakhani, 2010; Zaggl et al., 2023). Scholars have also begun to highlight how crowds *select* ideas, meaning that crowds also shoulder the effort of reducing the variation of ideas to focus attention on the most promising ones (Blohm et al., 2016; Csaszar, 2018; Hoornaert et al., 2017; Kornish and Ulrich, 2014; Mollick and Nanda, 2016). Using a similar approach as recent abductive machine-learning-informed work (e.g., He et al., 2020), we combine different methods to develop new insights into crowd selection. Table 3 summarizes our theoretical contributions, highlighting what we know from prior work and the novel theoretical insights emerging from our study.

6.1. Theoretical implications

6.1.1. Pruning explanations for crowd selection

Prior work investigating crowd selection has considered several ideator and idea characteristics that can explain crowd selection (Bayus, 2013; Deichmann and van den Ende, 2014; Riedl and Seidel, 2018; Szatmari et al., 2021; Beretta, 2019; Courtney et al., 2017; Li et al., 2016). However, these studies have focused on one or a few such factors, leading to many factors potentially affecting crowd selection and thus fragmented literature. Using a novel empirical approach of algorithm-based abduction on a large sample of ideas (He et al., 2020), we combine this fragmented literature to understand which of the explanations suggested by prior work matter most for crowd selection. This enables us to prune the literature by directing our joint attention to the most important explanations for crowd selection while disregarding those with less predictive power, which opens the door to building a more parsimonious theory.

It also highlights the benefits of combining methods to yield new theoretical insights (Shrestha et al., 2021). A deductive theory-building approach, for instance, would allow us to develop deep insights into a single predictor but would miss out on other, potentially more important, factors. Algorithm-based abduction can explain real-world phenomena by theorizing from generalizable patterns detected via machine learning (Shrestha et al., 2021). It can distinguish between pattern detection and pattern explanation, aspects typically entangled in prior research (He et al., 2020). By distinguishing between the two, we provide theoretical explanations for the observed patterns and theorize about new ones emerging through abductive reasoning. The approach can thus be pivotal for building and refining theories based on large amounts of data (von Krogh, 2018). While first studies are pioneering the approach (He et al., 2020; Shrestha et al., 2021), our paper is, to our knowledge, the first that used it to generate insights into crowdsourcing.

6.1.2. Weeding out bad ideas vs. picking winners

We further extend research on crowdsourcing and idea selection by investigating multi-stage contests and developing empirical insight and theory about what factors affect crowd selection at different stages. While we find support for some arguments provided by prior work (Bayus, 2013; Deichmann and van den Ende, 2014; Riedl and Seidel, 2018), our key finding is that many of the proposed factors (such as ideator status, past success, and idea presentation) only explain crowd selection at early stages, but less so at later stages.

Building on signaling theory, we highlight the role of signals for crowd selection and their dynamic effects across selection stages, as called for by recent studies (Moradi et al., 2023). We theorized that, at early stages, crowds rely on certain ideator and idea characteristics as signals for idea quality to single out low-quality ideas. Factors such as ideator status, past success, and a well-crafted idea presentation constitute costly signals that crowd members prioritize when weeding out ideas at early stages, weakening the effect of other low-cost signals. While we can predict crowd selection at early stages (and which factors matter the most), our findings show that predicting winners at the final stages is harder. We theorized that new signals emerge as ideas progress to later stages of selection and that these signals replace earlier signals. The early ideator- and idea-related signals fade. Passing earlier stages of crowd selection is a strong signal, which can trigger social influence processes and increase ideators' promotion efforts. In line with this, we identified *fast traction* as an important but often overlooked success factor in multi-stage crowd selection. Ideas that receive quick support have more chances to spread because such support may be perceived as a (very early) signal for idea quality, induce herding behaviors, increase visibility and promotion efforts, and generate word-of-mouth (on and off the platform). Therefore, it can lead to the cumulative advantage effect, as Watts and Gilbert (2011) discussed, with rich-getting-richer dynamics (Merton, 1968). Overall, a key contribution of this study is thus to *differentiate between weeding out weak ideas (at early stages) and picking*

Table 3
Comparing the novel insights of this study with prior work.

Main contribution	Known from prior work	Notable references	Novel insights emerging from this study
Pruning explanations for crowd selection	Prior work has focused on how different idea- and ideator-related factors affect crowd selection. These studies, however, have focused on studying only one or a few factors, and we know less about which ones matter the most.	Hoornaert et al. (2017); Szatmari et al. (2021); Jensen et al. (2014); Mollick and Nanda (2016); Piezunka and Dahlander (2015)	*Using the novel approach of algorithm-based abduction enables us to prune the literature and direct the attention to the most important explanations for crowd selection.* It has the advantage of distinguishing between pattern detection and pattern explanation to build and refine theories based on large amounts of data.
Weeding out bad ideas vs. picking winners	Prior work on single-stage settings highlights that we can explain which ideas crowds select as winners based on different idea- and ideator-related factors.	Blohm et al. (2016); Hoornaert et al. (2017); Szatmari et al. (2021); Jensen et al. (2014); Mollick and Nanda (2016); Piezunka and Dahlander (2015)	*Our study enables us to differentiate between weeding out bad ideas and picking winners in multi-stage selection environments.* We show that it is easier to predict which ideas crowds weed out at early stages of selection compared to final stages, despite the many predictors available from prior work. We theorize that this is because new signals emerge and replace ideator- and idea-related signals at later stages. *Fast traction can boost support at later stages of selection.*
	Ideator's status and experience matter to gain high levels of crowd support in single-stage settings.	Bayus (2013); Deichmann and van den Ende (2014); Schemmann et al. (2016); Riedl and Seidel (2018); Hoornaert et al. (2017); Li et al. (2016); Özaygen and Balagué (2018)	*Status and prior success experiences matter only for passing early stages of selection.* They act as early signals of idea quality that help crowds weed out ideas early in the process. However, status and prior success are less useful to predict winners at final stages. Indirect learning experiences gained by ideators are not an important predictor in multi-stage selection.
	Idea characteristics matter to gain high levels of crowd support in single-stage settings. Examples are the	Jensen et al. (2014); Mollick and Nanda (2016); Courtney et al. (2017); Hoornaert et al.	*These idea characteristics constitute important predictors only at early stages of selection*, while they

(*continued on next page*)

Table 3 (*continued*)

Main contribution	Known from prior work	Notable references	Novel insights emerging from this study
	length of the idea description, positive sentiment, and using media such as images.	(2017); Li et al. (2016)	have little bearing on crowd selection at final stages.
	Crowds can be leveraged to filter out bad ideas. However, empirical evidence is limited, mostly experimental, and based on crowds evaluating a low number of ideas. In these studies, crowds were asked to discern low-quality ideas. These studies do not explain why weak ideas may be discarded by crowds.	Toubia and Florès (2007); Klein and Garcia (2015); Cui et al. (2019)	By using the algorithm-based abductive approach on a *large sample of ideas*, we can explain *why weaker ideas are filtered out* by crowds at early stages and which factors matter the most for crowd selection.

winners (at final stages) in multi-stage crowd selection.

As such, our results complement earlier studies that used experimental methods and found that crowds can be leveraged to filter out bad ideas (Cui et al., 2019; Klein and Garcia, 2015; Toubia and Florès, 2007). In this work, crowds had to evaluate a few competing ideas and were explicitly instructed to identify low-quality ideas. However, these studies do not explain why crowds discard certain ideas and which factors affect this. We extend the findings from these studies by presenting a large-scale empirical investigation in a setting where many ideas compete across multiple stages. Our results suggest that crowds use ideator and idea characteristics as signals to filter out low-quality ideas.

Our insight into the difference between weeding out bad ideas versus picking winners links to recent discussions about the value of crowd selection (Dahlander et al., 2019). Organizations face a trade-off when involving crowds in selecting ideas, as such a decision entails advantages and disadvantages (Dahlander et al., 2019). One frequently discussed downside of crowd selection is that crowds do not necessarily select the best ideas due to social influence (Frey and van de Rijt, 2020; Hofstetter et al., 2018). Our results, however, allow a more positive interpretation. We had a hard time predicting winners at the final stages. At the same time, we know from LEGO that most of the successful ideas that were commercialized have become bestsellers, and many are currently sold out. This suggests that crowd selection brings value beyond weeding out weak ideas at early stages. At later stages, crowds may be able to detect small nuances in ideas. That is, they may be able to identify those ideas with "the certain something" that is hard to measure and thus that cannot be detected based solely on idea- and ideator-related factors. Similarly, in many settings, an idea's value can only be determined by the preferences of individuals and groups acting as selectors (Wijnberg and Gemser, 2000). Like in cultural markets (Becker, 2008), what is a good and novel idea is subjective and can, above a certain quality threshold, only be determined by what the crowd considers a good and novel idea. It is based on socially constructed judgments. In line with this, Osterloh and Frey (2019) argue that the unpredictability of outcomes should not necessarily be seen as a nuisance to be eliminated. Instead, randomness can be leveraged purposefully as a decision-making mechanism to allow for the detection of good ideas (Osterloh and Frey, 2019). Recent debates have similarly highlighted that social influence may accelerate the discovery of good content (Celis et al., 2016). That said, LEGO Ideas combines crowd and organizational selection, and the company's experts likely play an important role in its success. Still, the involvement of crowds across multiple selection stages has been valuable for the company as it enables the detection of high-quality ideas with strong market demand.

6.2. Managerial implications

Asking crowds to generate ideas often overwhelms managers with more options than they can implement (Park et al., forthcoming; Piezunka and Dahlander, 2015). One potential solution is involving crowds in selecting ideas across multiple stages. Our work suggests that ML algorithms can predict which ideas are weeded out at the early stages of crowd selection but that picking winners at later stages is less predictable. This implies that organizations could use algorithms to pre-filter early ideas. Beyond saving resources, automatically filtering low-quality (or even counterproductive) ideas can engage crowds because it saves their attention for high-quality ideas. A higher proportion of high-quality ideas makes the platform more attractive and may prompt longer and more frequent visits. It may also stimulate user-to-user interaction, generating more value from the platform allowing them to learn (see also Gaessler and Piezunka, forthcoming). Involving crowds can bring unique value not by weeding out bad ideas but by selecting ideas with that certain something. Our study also offers managers practical insights into achieving crowd support. Managers may want to leverage the role of early crowd support. Platform designs that reward trending ideas with prominent display positions and additional attention may not always be in the organization's best interest as they lead to rich-getting-richer dynamics. At the same time, platform managers may want to encourage ideators to invest in their platform presence and idea presentation as these (costly) signals help to achieve (early) crowd support.

6.3. Limitations and directions for future research

This paper relied on rich quantitative and qualitative data from the LEGO Ideas platform. While the deep access enabled us to combine multiple methods, it also means that our findings, as with any single-firm study, may be domain specific. Our theory, therefore, has boundary conditions. First, it is developed by studying an open platform (everyone can take part in LEGO Ideas), entirely virtual, and relies on the self-selection of crowds in voting. Unlike other idea competitions, the projects on LEGO Ideas are outcome prototypes. Future research should therefore investigate whether our findings apply to other crowdsourcing settings; for instance, closed crowdsourcing platforms that only invite certain individuals to participate or platforms focusing on soliciting ideas rather than outcome prototypes. Second, LEGO Ideas has a mixed incentive structure that provides recognition through different gamification principles, various prizes, and monetary rewards. Such a mixed incentive structure may influence the crowd members' voting at different stages. This calls for research that explores how crowds evaluate ideas in settings with different incentive systems. We recognize that it is difficult to discern whether these attributes of the empirical context make some explanations provided by prior work less critical in our setting. An ultimate answer would require studies that compare crowd selection on different platforms and test the effect of various design choices.

Finally, our empirical approach has drawbacks. While abductive theory-building based on ML has many merits (on which we elaborated), it can be prone to new biases caused by (limited) data availability. Just as studies attempting to build theory from survey observations are subject to sampling biases, abductive theory-building may be subject to missing variable biases because it is limited to the included variables. Missing variable biases are not, per se, an artifact of ML. As ML techniques are typically applied to large-scale observational data, as in our study, the included variables—or conversely, the missing variables—are

the Achilles' heel of abductive theory-building. Researchers naturally attempt to build theories on the effects consistently present across well-performing models, but these models are blind to omitted variables.

Acknowledging the boundary conditions mentioned above, our findings likely generalize to other crowdsourcing platforms relying on multiple stages of crowd selection. We are particularly optimistic about crowdsourcing platforms where the number of votes and others' ideas and comments are visible to the crowd members. We also believe that our findings are relevant beyond firm-hosted crowdsourcing platforms. For instance, we increasingly observe the involvement of citizens in government open crowdsourcing initiatives. These initiatives typically involve multiple rounds of voting. Similarly, many democratic online petitions entail multiple voting rounds before the petition is considered by state representatives (Chen et al., 2019). Our findings suggest that having built a high status and investing efforts in developing a well-crafted petition likely matter most for the early stages of the petition. Later, such initiator and petition characteristics may be outweighed by other, less predictable factors such as social influence or campaigning—just as in our setting.

7. Conclusion

Using rich data from LEGO Ideas and a novel approach of algorithm-based theory building, we provide new insights into crowd selection in multi-stage contests. Our key finding is that certain characteristics of the ideator and idea are useful to predict which ideas are weeded out at early stages but much less to predict which ideas are selected as winners at final stages.

CRediT authorship contribution statement

LD, MB, AT and LF contributed equally and are first authors.

Declaration of competing interest

We declare no conflict of interest. The data comes from a long collaboration with LEGO. We have no financial interest in LEGO. They are simply research partners. The data is unique and there is no other published data from this collaboration. We cannot disclose the data as it is proprietary. We would like, however, to disclose all code we have used as we have developed many new things for other researchers to reuse.

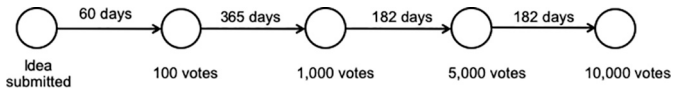
Data availability

The data that has been used is confidential.

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Appendix 1. Overview of multi-stage crowd selection on LEGO Ideas



Appendix 2. Emergent themes and variable definitions for ML

Emergent themes from qualitative coding	Definition and construction of variables for machine learning
Ideator attributes	
Tenure	Number of months between the user profile was created and project submission.
Age	Age of the ideator measured in years.
Gender	Gender of the ideator (coded as 1 for men, and 2 for women).
Ideator experience	
Projects launched	Number of projects launched before the submission of the focal idea.
Previous success	Dummy variable that indicates whether the ideator obtained at least 100 supporters for a project before the submission of the focal idea.
Logins	Number of logins up before submission of the focal idea.
Votes cast	Number of supports the ideator has given before submission of the focal idea.
Comments written	Number of comments written on others' ideas before submission of the focal idea.
Users followed	Number of users the ideator was following before submission of the focal idea.
Sentiment of comments	We apply five different dictionaries for sentiment analysis (Go et al. 2009, Hu and Liu 2004, Mohammad et al. 2013, Mohammad and Yang 2011, Wilson et al. 2005). These dictionaries represent five broadly adopted computer translations between word and phrase usage and positivity. To limit the potential bias from using a particular dictionary, we standardized and averaged the scores across these five dictionaries into one measure. This measure captures positivity in the comments.
Variation in sentiment	The average of the standard deviations of *Sentiment of comments* (based on the five sentiment dictionaries). We follow the same procedure as for *Sentiment of comments* to obtain a robust measure.
Ideator status	
Total votes received	Number of supports the ideator received before submission of the focal idea.
Highest number of votes received	Maximum number of supports the ideator received on a single idea before submission of the focal idea.
Personal followers	Maximum number of followers the ideator has had on the submission dates of former projects.
Unique followers	Number of users who were following the ideator before submission of the focal idea.
Comments received	Number of comments received before submission of the focal idea.
Sentiment of comments received	Mean sentiment of comments received before submission of the focal idea.
Variation in sentiment received	Average of the standard deviations of *Sentiment of comments received* (based on the five sentiment dictionaries).

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Emergent themes from qualitative coding	Definition and construction of variables for machine learning
Idea topics	
Idea topics	We used LDA topic modeling to search for latent structures in the co-occurrences of words. We applied unsupervised text mining techniques to free text and users' own idea classifications, potentially identifying the projects' cross-characteristics. The input text for this analysis was a single text vector for each project, containing its title, description, and all metadata tags. We derived 15 latent text topics, and we named the topics qualitatively according to the primary words associated with the latent topics. These were the topics: colors, game, fame, world, city, details, vessel, fantasy, vehicle, figure, hero, nature, building, war, and space.
Number of tags	Number of tags assigned to the idea.
Idea image characteristics	
Number of images	Number of images uploaded with the idea. It is the most basic visual metric and may be correlated with multiple characteristics of the idea (Li et al., 2016), such as the level of detail, how excited the ideator is about the idea, and the bandwidth for image uploading available to the ideator.
Size of the principal image	The original size of the principal image (in kilobytes). Each idea can have multiple images, and one of them serves as the principal image that identifies the idea on the front page. The file size of the principal image may be correlated with two characteristics of the idea: (a) camera/smartphone quality, acting as a proxy for the ideator's financial means, and (b) commitment to the idea. For example, digital demo models require less computer memory than high-resolution professional images of built demo models. Digital demo models may also be associated with limited financial ability to purchase LEGO building blocks but may be cognitively expensive for ideators with low IT affinity. Image size thus must be interpreted carefully.
Colorfulness	As a measure of the colorfulness and creativity, we calculated a measure for the diversity of color usage in the idea's principal image with the OpenCV python package (Hasler and Suesstrunk, 2003). Diversity of color usage captures salience of an image, and salience attracts attention (Hays et al., 1972). At the same time, it is important to acknowledge that color use varies with themes, e.g., the arctic ice research outpost is mostly white, and the amusement park has more colors.
Complexity	We measured the complexity by calculating the principal image's relation between file size (in kilobytes) to area (in square-inches). All image files are JPG files, which is a file format that automatically minimizes the data per area based on color similarity within images (Ahmed et al., 1974, Pennebaker and Mitchell, 1992). The measure is therefore a proxy for idea complexity, as the increased complexity of ideas makes images harder to reduce. We note that LEGO requires images to be on clean backgrounds, to not confound idea proposals with idea framing.
Idea text characteristics	
Length of description	Number of words in the description of the idea.
Sentiment of description	We scored the idea description in terms of its sentiment (based on five sentiment dictionaries). We followed the same method as in *Sentiment of comments* and the other sentiment measures.
Resubmission count	Number of idea resubmissions before the submission of the focal idea.

Appendix 3. Synthetic Minority Oversampling Technique (SMOTE)

The predictive power of ML models in classification problems decreases if the proportion of data points belonging to the minority class is low (imbalanced data). One way to address this issue and arrive at a more balanced dataset is to oversample the minority class by synthesizing new data points based on existing ones. There are many ways of oversampling the minority class and one of the most prominent ones is the Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al., 2002). We used the SMOTE function in the *imblearn* package in Python to generate synthetic data points from our training data sample.

Appendix 4. Time-series model to check stationarity

We tested the stationarity of our data between 2016 and 2018 using the Augmented Dickey-Fuller (ADF) test. The ADF test is a unit root test to check if there is serial correlation (with time dimension, i.e., created date of the project). We used the *statsmodels* package in Python and its *adfuller* function to perform the test. Results show that there is no unit root in our features (with 95 % significant level), confirming that our data are stationary. The adfuller function returned the following values:

Variable	ADF statistic
Idea reaching 100 votes	−80.48***
Idea reaching 1000 votes	−100.27***
Idea reaching 5000 votes	−204.02***
Idea reaching 10,000 votes	−50.13***
Tenure	−17.33***
Age	−25.22***
Gender	−34.63***
Projects launched	−26.87***
Previous success	−29.15***
Logins	−28.17***
Votes cast	−27.84***
Comments written	−23.23***
Users followed	−23.95***
Sentiment of comments	−26.44***
Variation in sentiment	−23.48***
Total votes received	−87.57***
Highest number of votes received	−72.02***
Personal followers	−23.20***
Unique followers	−39.64***
Comments received	−26.04***

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Variable	ADF statistic
Sentiment of comments received	−21.45***
Variation in sentiment received	−21.46***
Number of tags	−74.52***
Number of images	−49.48***
Size of the principal image	−14.00***
Colorfulness	−98.40***
Complexity	−88.21***
Length of description	−27.14***
Sentiment of description	−98.31***
Resubmission count	−45.327742***

*** $p < .001$.

Appendix 5. Comparing ML and random guessing

Two-sample t-test

To check whether our ML models improved random guessing significantly, we used test of equal or given proportions. The test clears whether or not two independent samples have the same proportions of a class. For example, to be able to check whether the recall measure of our model can perform significantly better than random guessing, we have to compare the proportion of true positives in the whole test sample with the proportion of true positives among true positives + false negatives in the result of performing the model on the test sample. We used the *prop.test* function in the *stats* library of R. The test shows that for all precision, recall, and F1 measures, both of our algorithms (LASSO and GB) perform better than random guessing.

McNemar's test

When comparing two different algorithms in a classification problem, one must check whether one algorithm is significantly better than the other. Each algorithm will be run on the test sample and then a contingency table is computed that contains the number of true and false predictions for each method (true-true, true-false, false-true, and false-false). The *p*-value of the statistical test (H_0 = similar methods; H_1 = different methods) identifies if we can reject the null hypothesis. In our research, we used the *statsmodels* package in Python to confirm that both LASSO and GB significantly improved the performance of random guessing.

Appendix 6. Precision and Recall

There are different measures of performance of ML models in classification problems. The objective of the research design and the data structure are the most important factors in selecting the most appropriate and relevant measure of model performance. Since our research focuses on discovering the ideas selected at each stage, we picked the measures based on understanding and relevance of the specific (rare) class (success). The precision and recall measures align with this objective of focusing on true positives. These measures are calculated as:

$$\text{Precision} = \frac{True\ Positives}{True\ Positives + False\ Positives} \text{Recall} = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

Precision focuses on not considering failures as successes and recall focuses on not missing successes. As our dataset is highly imbalanced, not missing successes is our priority. We therefore used recall as our main measure. However, we also computed F1 and Matthews correlation coefficient (MCC) measures as a robustness check, combining precision and recall measures. F1 is a harmonic mean of precision and recall, which means both measures have the same weight when calculating the F1. F1 performs better when both precision and recall are close to each other, whereas MCC is suitable when the dataset is highly imbalanced in terms of statistical rate.

Another measure based on precision and recall is the Area Under Precision-Recall Curve (AUPRC). It sets the number of steps (equal to the number of subsamples) and calculates the precision and recall for each step (threshold) to create pairs of points (precision and recall) to draw the plot. The area under this curve is calculated as a combined measure of finding true positives among false positives and negatives.

Appendix 7. Cross-validation

Cross-validation is a statistical method used to estimate the power of ML models. Each ML method is built on two main steps—training and testing. Cross-validation implements the test step inside the training step. The method divides a limited training data sample into *k* separate folds by randomly assigning each data point to one subsample. Then, for *k* times, it sets one of the folds (usually the first one) as its target (test sample) and trains on other folds. The main goal of cross-validation is to estimate the power of a ML model to predict new unseen data. The second goal is to tune hyper-parameters (the parameter values are going to be set before or during the training step) for some ML models that have these parameters in their algorithms. In each iteration, hyperparameters of the model would be tuned to enhance the desired performance measure (in our case, recall).

Appendix 8. Ranking variables through GB and tuning parameters in GB

Extreme Gradient Boosting

When ML methods rank the variables based on their importance for improving model performance, the correlations among variables cause an underestimation of the importance of variables that are correlated to the selected ones. This is because ML methods cannot improve their performance

by using correlated variables as much as they can by using uncorrelated ones. To arrive at a correctly ordered table of the most important variables, we first ran the ML algorithm on all our variables, then dropped the most important ones and reran the algorithm.

Extreme Gradient Boosting (XGBoost) belongs to a family of boosting algorithms that can convert weak learners (which are only slightly better than random guessing) into strong learners. Boosting iterates growing a decision tree (as a classifier) based on a subsample of the training dataset, but with a focus on the data points that were misclassified in the previous step. By focusing on the errors from the previous step, a better decision tree—in terms of classification performance—is built. We used the *mlr* library and the *xgboost* package in R to implement XGBoost.

Tuning parameters

We tuned the parameters in each step to have the best performance in terms of the model's precision. The parameters we tuned to train the model were the maximum number of iterations (*nrounds*) and the depth of the decision tree (*max_depth*). We started with all data points and all features in each step. We selected the most important feature in each step, filtered it out, and then reran the algorithm based on the remaining features. In this way, we also counted the features correlated with the most important feature we filtered in the previous iteration, for which the ranking score would otherwise be diminished.

Ranking of variables

The table below shows the ranking of variables, as generated by the *xgboost* package. For each variable, the package computes a Feature Share in the built model. As we just picked up the first feature in each step (the variable with the highest feature share) and reran the model, we could not compare the Feature Shares for various variables directly. Therefore, we considered AUPRC of the model as the Model Score (using the performance of the model on the test data) and multiplied the Feature Share with the Model Score to calculate the final Feature Score. It shows the weight of the variable in idea success—as it is our goal. The final ranking is based on the Feature Score.

Relative variable importance in GB model for explaining an idea's success (based on recall measure)

Variable (target: Reaching 100 votes)	Feature share	Model score	Feature score
Length of project description	46.04	83.75	38.56
Resubmission count	48.41	69.82	33.80
Previous success	43.98	66.83	29.40
Users followed	39.60	70.81	28.04
Number of tags	34.74	72.47	25.17
Number of images	28.70	71.81	20.61

Variable (reaching 1000 votes)	Feature share	Model score	Feature score
Number of images	32.28	73.80	23.82
Previous success	33.80	66.67	22.53

Variable (reaching 5000 votes)	Feature share	Model score	Feature score
Number of tags	39.99	78.26	31.29
Length of project description	41.77	73.91	30.87
Number of images	34.46	65.22	22.47
Unique followers	26.43	65.22	17.23

Variable (reaching 10,000 votes)	Feature share	Model score	Feature score
Length of project description	29.25	68.75	20.11
Age	30.64	62.50	19.15
Highest number of votes received	23.00	75.00	17.25

Notes: Feature Share, Model Score, and Feature Score are percentages. Feature Share is the degree of feature's importance in the creation of the model. The scikit-learn package in Python generates these percentages for all features it uses to create the model. Model Score is the recall for the test dataset. Feature Score is Feature Share × Model Score that means how much is the weight of the variable in improving the performance measure (recall) based on test dataset. We report features for which the recall measure dropped when we dropped them from the set of features used to create the model.

Appendix 9. LASSO performing feature selection based on the F1 measure

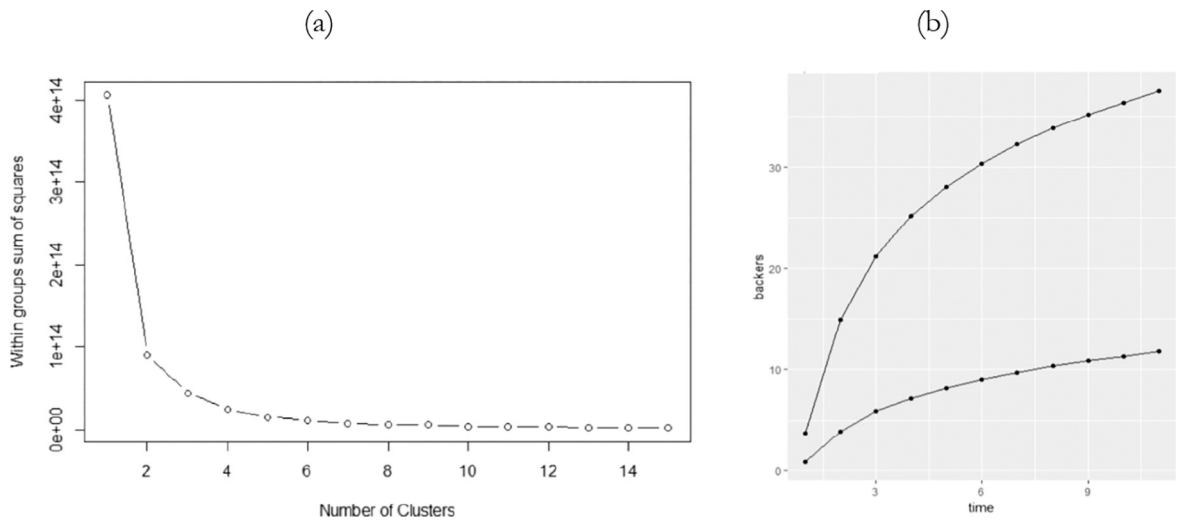
Variables	Predicting 100 votes	Predicting 1000 votes	Predicting 5000 votes	Predicting 10,000 votes
Ideator attributes				
Tenure	1.66	1.84		
Age	2.27	2.24	2.10	
Gender		−0.92	−1.46	
Ideator experience				
Projects launched	−5.26	−7.92	−3.22	−7.80
Previous success	1.69	2.39	0.21	

(*continued on next page*)

(*continued*)

Variables	Predicting 100 votes	Predicting 1000 votes	Predicting 5000 votes	Predicting 10,000 votes
Logins		−1.17		
Votes cast		−2.52		
Comments written				
Users followed				
Sentiment of comments		−3.78		
Variation in sentiment	−2.23	−2.73		
Ideator status				
Total votes received				
Highest number of votes received		4.26	3.67	2.93
Personal followers				
Unique followers				
Comments received				
Sentiment of comments received				
Variation in sentiment received		−4.76	−0.52	−1.33
Idea topic				
—building				
—city				
—color				
—details				
—fame				
—fantasy				
—figure				
—game				
—hero				
—nature				
—space				
—vehicle				
—vessel				
—war				
—world				
Number of tags	10.08	10.10	4.65	6.84
Idea image characteristics				
Number of images		3.88		
Size of the principal image		1.21	1.64	
Colorfulness		−0.79		
Complexity				
Idea text characteristics				
Length of description	6.29	6.25	9.17	7.53
Sentiment of description	5.45		5.46	4.49
Resubmission count	−5.70	−5.24	−3.58	
Trends				
Day of the week dummies				
Month of the year dummies				
Observations				
Observations for training	30,272	30,272	30,272	30,272
Observations for testing	7568	7568	7568	7568
Model fit				
Baseline (random guessing) (%)	7.96	1.28	0.29	0.20
Precision (%)	30.35	7.78	2.54	1.87
Recall (%)	76.95	74.23	75.00	81.25
F1 (%)	43.53	14.08	4.92	3.66
MCC (%)	41.58	21.56	12.71	11.50
True positives	464	72	18	13
True negatives	5900	6617	6855	6871
False positives	1065	854	690	681
False negatives	139	25	6	3

Appendix 10. How early support predicts future support: cluster analysis



Notes: Figure (a) shows how cluster heterogeneity decreases dramatically when splitting all ideas into two pools of similar projects in terms of votes obtained within the first 60 days. This chart shows that introducing additional clusters reduces within-cluster variation only to a much lesser degree, which can be seen from the flattening curve. Figure (b) profiles the two clusters in terms of how many supporters they have obtained over their first weeks on the platform on average, and we note that successful ideas stand out from the start.

Appendix 11. Testing H1 and H2: logit regressions on Sample II

	Model 1 (logit regression on 100 votes)	Model 2 (rare event logit on 10,000 votes)	Model 3 (rare event logit on 10,000 votes)
	H1	H1	H2
Projects launched	−0.0281*** (0.0064)	−0.0267 (0.0301)	−0.04397 (0.0341)
Previous success	1.685*** (0.178)	0.576 (0.854)	−0.6168 (1.084)
Total votes received	0.0001*** (0.00002)	0.00008*** (0.00003)	0.00001 (0.00002)
Number of tags	0.0186** (0.009)	0.0279** (0.0108)	0.116*** (0.034)
Number of images	0.0145** (0.006)	0.0295^ (0.0171)	−0.0416 (0.082)
Length of description	0.0006*** (0.00009)	0.0004^ (0.0002)	0.0002 (0.0003)
Sentiment of description	0.0397*** (0.0101)	0.0349 (0.0396)	0.0697 (0.0499)
H2: Fast traction			−2.636*** (0.604)
Constant	−3.8296*** (0.112)	−7.45*** (0.545)	−1.458 (0.979)
Wald-Chi square	355.26	58.32	32.40
Log-likelihood	−873.034	−11.305	−18.182
Observations	4205	4205	335

Note: Two-tailed tests.
^ $p < .10$.
** $p < .05$.
*** $p < .01$.

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